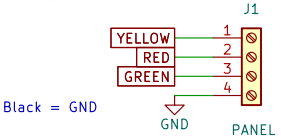
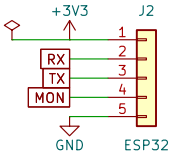
ECP (Honeywell/Ademco) interface – BJT version

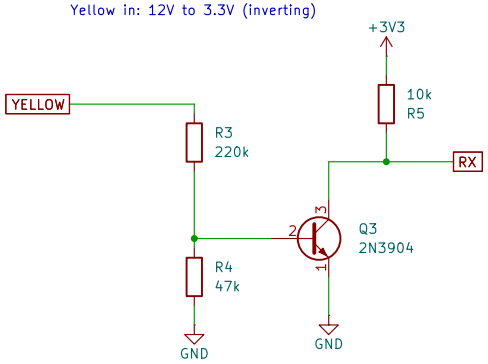
J1: 4–position 5.08 mm screw terminal, 22–16 AWG alarm wire



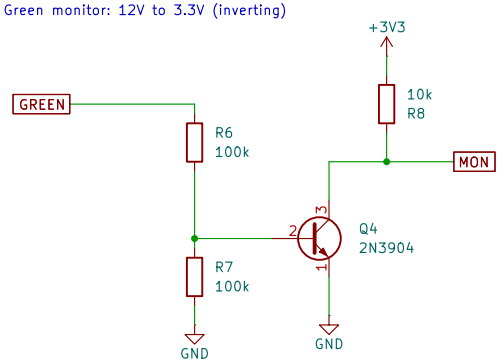
Black = GND



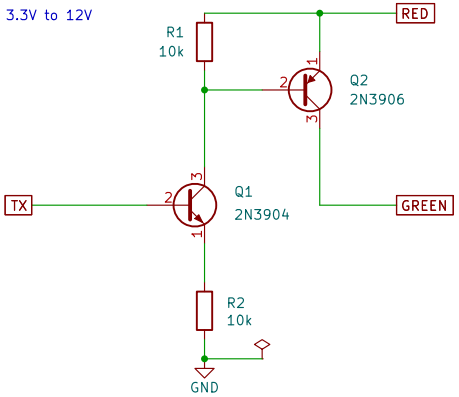
Yellow in: 12V to 3.3V (inverting)

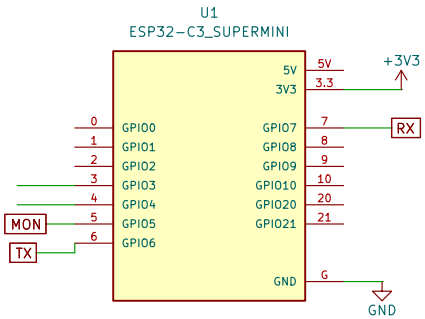


Green monitor: 12V to 3.3V (inverting)



Green out: 3.3V to 12V





YAML config:

```
  invert_mon: "false"
  invert_rx: "false"
  invert_tx: "true"
  input_mode_rx: INPUT
  input_mode_mon: INPUT
```

Note: Since the internal logic already inverts the incoming data the above settings will be the reverse to what the circuit does.

Suggested pins for RX,TX and MON
ESP32: 25/26/27 C3,S3,S2: 4/5/6 ESP8266: D1/D2/D5

ESP32C3 supermini pins:
rxpin: 7
txpin: 6
monpin: 5

Seeed XIAO ESP32C3 pins:
rxpin: 4
txpin: 3
monpin: 2

ESP32S3 supermini pins:
rxpin: 43
txpin: 44
monpin: 1

Seeed XIAO ESP32S3 pins:
rxpin: 3
txpin: 2
monpin: 1

Notes:
i: resistors r5,r8 and the 3.3v connection can be omitted but you will need to set:
   input_mode_rx: INPUT_PULLUP
   input_mode_mon: INPUT_PULLUP
ii: q1,q2,q3 and q4 can be substituted for other common small signal transistors.
iii: you can substitute resistor values but keep it within about 10% of the value up or down. You may go down to 4.7k for all 10k resistors.
iv: Q2 2N3906 PNP, Q1/Q3/Q4 2N3904 NPN. TO–92 pin order E–B–C assumed, check datasheet.
v: RX and MON are pulled up to the ESP32 3.3V by R5 and R8.
   Yellow idles high (panel 10k pull–up, low ~0.7V).
   Green idles low (panel 10k pull–down, 0V).
v: board can be mounted with standoffs using the 4 holes or using
   the 4 support pins on each corner